

COGNITIVE ROBOTICS COURSE PROJECT

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Abstract—This technical report provides an outline of how the tasks were implemented for the Cognitive Robotics course project using Kinova Gen3 7DOF robot arm. The project implements a learning from demonstrations i.e. Imitation Learning which is further finetuned by using Reinforcement Learning and evaluates the outcomes of using these 2 approaches together on a kitchen manipulation task. We set up a custom scene resembling a part of our kitchen in lab C069 in Isaac lab. We used this scene to collect data in simulation. For our first approach, we trained a behaviour cloning agent for imitation learning which was further finetuned using PPO for Reinforcement Learning. Testing was conducted on the Kinova arm, demonstrating robust performance for this combined approach.

Index Terms—Imitation Learning, Reinforcement Learning, Behaviour Cloning, PPO, Evaluation, Isaac Lab Custom Scene

I. INTRODUCTION

A. Motivation

This project addresses the challenge of enabling robots to efficiently learn complex manipulation skills from human demonstrations. Current robotic learning often requires extensive programming or vast trial-and-error. We aim to investigate and combine two advanced robot learning approaches on a Kinova Gen3 robotic arm for the below tasks trained using our own dataset:-

Task: Navigate to Fridge - Pick an object from Fridge - Navigate to Microwave - Place it in Microwave

(Simulation assets have been used from lightwheel which can be accessed at <https://www.lightwheel.ai>)

B. Problem Statement

The below approach has been implemented:-

- 1) Imitation Learning for the initial policy before performing RL.
- 2) Reinforcement Learning (RL) where the learning process is kick-started by an initial policy derived from direct imitation of human actions

This problem is deeply related to cognitive robotics as it explores how artificial agents can acquire and generalize skills through observation, a fundamental aspect of human cognition. Specifically, it delves into:

- Learning from Demonstration: A core cognitive ability to learn by watching demonstrations.
- Efficient Skill Acquisition: Key objectives in creating intelligent robots that can adapt like cognitive systems.

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†Supervised by Dr. Aleksandar Mitrevski

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By comparing these methods, the project aims to provide insights into developing more capable and intuitive learning mechanisms for robots, allowing them to better understand and replicate human-demonstrated tasks.

C. Proposed Approach

We focus on the cognitive robotics aspect of learning from demonstration (LfD).

- Reinforcement Learning-based Manipulation: Using Behavior Cloning for initializing RL policy using imitation learning and reward shaping to train a RL policy using Proximal Policy Optimization that mimics expert actions.

This report details the implementation of each component, including custom scene setup in Isaac Lab, the algorithms used, integration approaches, and performance evaluation on the robot. The system was tested in simulation using Isaac Lab. The report is divided into two main sections :-

- Section 2 - Custom scene setup using Isaac Lab
- Section 3 - Dataset collection in simulation
- Section 4 - Behaviour Cloning based Imitation Learning + PPO based RL

RELATED WORK

This work is based on the following prior work done in the field :-

The overall implementation leverages Isaac Lab for simulating our custom kitchen scene. Isaac Lab was introduced as ORBIT in the paper by Mittal et al. [1] The behaviour cloning (BC) algorithm was introduced by Bain et al. [2]. The imitation library helped us easily understand the BC algorithm better. [3] The imitation learning and reinforcement learning approaches were discussed in brief during the MAS Cognitive Robotics course [4] and is what inspired this project.

II. CUSTOM SCENE IN ISAAC LAB

We have used Isaac Lab simulator in order to create our custom kitchen scene which represents the relevant parts of the C069 Labs kitchen such as the shelf, microwave and fridge along with the Kinova Gen3 arm with Robotiq gripper in simulation. Setting up this scene was not straightforward and had a long learning curve involved. We watched a lot of tutorial videos on this and finally managed to find relevant usd files for the components in our scene from lightwheel. These assets were then combined in our simulation setup to complete our scene which involved a lot of trial and error. We also had to

figure out how to use the IK for the arm. This again was time consuming as there was no straightforward guide available on this. But finally we managed to set it up too. One of the biggest issues we were facing was that attaching the gripper and camera to the arm was not easy and failing a lot, but we managed to resolve it.

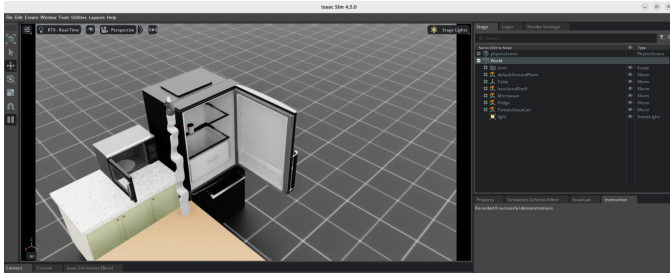


Fig. 1: Custom Kitchen scene in Isaac Lab

III. DATASET COLLECTION IN SIMULATION

We are using the same custom kitchen scene in Isaac Lab to collect data using tele-operation using our custom tele-operation script adapted for our arm based on Isaac Labs inbuilt teleop capabilities. We used our keyboard for moving and manipulating the arm and the gripper. Collecting this data was not easy as teleoperation in simulation using only keyboard is not as intuitive for manipulation tasks. We were still able to collect a dataset for our task and stored the demonstrations as hdf5 files.

IV. IMITATION AND REINFORCEMENT LEARNING IN ISAACLAB

For our experiments we employed two complementary learning approaches: *Behaviour Cloning (BC)* for imitation learning and *Proximal Policy Optimization (PPO)* for reinforcement learning. Both were implemented within the IsaacLab framework, which provides built-in integration with robomimic and skrl respectively.

A. Behaviour Cloning with Robomimic

Expert demonstrations were collected via teleoperation in simulation and stored in HDF5 format. To enhance data efficiency, IsaacLab's Mimic module was used to generate additional synthetic demonstrations. The behavioural cloning agent was trained using Robomimic through the following interface:

```
./isaacclab.sh -p scripts/
imitation_learning/robomimic/train.py
\
--task <TASK_NAME>
--algo bc
--dataset <path/to/dataset.hdf5>
```

Model checkpoints were periodically stored during training (e.g., every 100 epochs). Evaluation was performed using

the `play.py` script, which executes the learned policy in simulation for a fixed number of rollouts:

```
./isaacclab.sh -p scripts/
imitation_learning/robomimic/play.py
\
--task
    <TASK_NAME>
--num_rollouts 50 \
--checkpoint
    <path/to/model_checkpoint.pth>
```

This process enables direct assessment of how closely the imitation policy reproduces expert demonstrations.

B. Proximal Policy Optimization with skrl

Using our BC agent, we further finetuned it with reinforcement learning, we used the built-in integration of IsaacLab with the `skrl` library. PPO, a widely used on-policy policy gradient method, was employed to optimize the control policy directly from interaction with the environment. Training was launched using the following command:

```
./isaacclab.sh -p
scripts/reinforcement_learning/skrl/train.py
--task <TASK_NAME> --headless
```

During training, the agent continuously interacts with the simulated environment, collecting rollouts and updating its policy parameters according to the clipped surrogate objective of PPO. Periodic checkpoints were generated and could be evaluated with the corresponding play script:

```
./isaacclab.sh -p
scripts/reinforcement_learning/skrl/play.py
--task <TASK_NAME>
--num_envs 32
--use_pretrained_checkpoint
```

This enabled quantitative and qualitative evaluation of the learned policy under the same conditions as imitation learning.

C. Comparison

By training policies using imitation (BC) and further fine-tuning it with reinforcement (PPO) learning, we were able to compare how efficiently the approach adapted to the given task. While BC provides a fast initialization from expert trajectories, PPO enables fine-tuning through direct environment interaction.

V. RESULTS & EVALUATION

First, We tested both our approaches in simulation on the Kinova Gen3 robot arm in Isaac Lab, using using our custom kitchen scene where the policy ran successfully.

The policy when deployed on the robot worked sufficiently well to for our approach, correctly meeting expectations as defined in the problem statement.

A. Behaviour Cloning for Imitation Learning RL

The trained BC agent was not up to the mark on its own and needed further finetuning using reinforcement learning. Our PPO based approach helped the policy solve our task.

Generalization Capabilities :- Environmental Changes: Vary bin positions, lighting conditions, and workspace configurations

Training Efficiency :-

Low number of demonstrations needed for training

Assessment by a human:-

Naturalness Scoring: How human-like the robot movements appear on a scale of 1-10 i.e. we would rate it a 7

Safety Evaluation: Avoids collisions and gently handles object

Trajectory : BC agent has not as smooth trajectory but it gets better post RL training

Time to train : BC training took less time and RL took more time

VI. CONCLUSIONS

A. Summary

Based on testing on the Kinova robot arm in the real world, we conclude that the solution for behaviour cloning has been successfully implemented and further used to train our RL agent successfully as well. The system reliably meets the project's objectives by performing the task sufficiently for our approach which we were further able to evaluate. Overall, while we reduced the scope of having 2 different tasks as specified in the project proposal which was a bad decision by us considering the time available and this certainly taught us a valuable lesson in time management, we were able to successfully demonstrate our approach for one of the task and we believe this is sufficient to satisfy our project goals for now. In the future, we hope to further test it against new tasks and also use other approaches including those defined in our original scope.

B. Contributions

Sai Mukkundan - Custom kitchen scene setup in Isaac Lab simulation, Behaviour Cloning for Imitation Learning

Aaron Cuthinho - Teleoperation in simulation and data collection, PPO for Reinforcement Learning

Saloni Pathak - Setting up of a Robocasa custom kitchen scene for testing purposes which did not end up working for data collection, Report Editing

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ACKNOWLEDGMENT

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STATEMENT OF ORIGINALITY

We, the undersigned below, declare that this work has not previously been submitted to this or any other university and that it is, unless otherwise stated, entirely our own work. The report was, in part, written with the help of the Github Copilot (Gemini 2.5 Pro model) as described in the appendix. We are aware that content generated by AI systems is no substitute for careful scientific work, which is why all AI-generated content has been critically reviewed by us, and we take full responsibility for it.

August 31, 2025

Date

Aaron Cuthinho

Signature

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Sai Mukkundan

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APPENDIX

C. Link to the Code Repository

The code implementation for the project is available in the following Github repository :-

- https://github.com/saiga006/GenReal_CogRob

D. Description of AI Use

Github Copilot with the Gemini 2.5 Pro model has been utilized to partly(very little use) write this report, such as automatic generation of some text, and in conversion of the report's content to the Latex format.

REFERENCES

- [1] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar, A. Mandelkar, B. Babich, G. State, M. Hutter, and A. Garg, "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3740–3747, 2023.